

Anglo-Irish natural philosopher, theologian, and writer Robert Boyle carried out innovative experiments on the nature of gases. Widely considered the first modern chemist, he redefined chemistry as a discipline and broke away from the Aristotelian methods that had previously governed scientific endeavor.

Born into a wealthy Irish aristocratic family, Robert Boyle was educated at Eton College and continued his studies during an educational tour of Europe. In Italy, he witnessed the reactions to the death of astronomer Galileo Galilei (see pp.50–55) and became greatly inspired by the idea of scientific discovery. In 1644, he moved to England and, although a devoutly religious man, pursued a career in experimental science that epitomized the 17th century's shift towards rationalism.

New era

During the early 1600s, scientific study was typically based on the principles outlined by Aristotle 2,000 years before (see pp.14–17). However, a new era was dawning in which theories were being rigorously tested by experimentation. In London, Boyle met a group of intellectuals who shared his passion for science, which at the time was known as “natural philosophy.” This group, called the Experimental Philosophy Club, bought all the equipment they needed to perform experiments and shared their results with

To test whether sound can pass through a vacuum, Boyle placed a bell in a container and drew the air out. As the vacuum increased, the sound of the bell grew fainter.



MILESTONES

NEW PHILOSOPHY

Settles in England in 1644, and turns to experimental science; becomes the leading figure in the field.

AIR PRESSURE

Designs an air pump with Robert Hooke in 1659; pens his first scientific paper on air pressure in 1660.

UNIVERSAL MATTER

Publishes *The Sceptical Chymist* in 1661, which rejects Aristotle's account of the elements.

BOYLE'S LAW

Establishes the universal law of air pressure (the first gas law), publishing his results in 1662.

“Even when we find **not** what **we seek**, we find **something else.**”

Robert Boyle, 1661